Store Inventory Positioning, E-Commerce Intensity, and Retailer Profitability: Evidence From U.S. Census Microdata

by

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Abstract

Problem definition: Retailers increasingly sell both in physical stores and via e-commerce channels. In this setting, a central yet underexplored strategic decision concerns store inventory positioning (SIP)—the share of total inventory held in stores rather than in upstream facilities—and its effect on firm performance. While classical inventory theory highlights a cost–responsiveness tradeoff associated with SIP, little empirical evidence exists on how SIP affects retailer profitability in a multi-channel context. This study investigates how SIP relates to operating margins and how this relationship varies with retailers' e-commerce intensity (ECI), the share of total sales facilitated via online channels. We also explore whether these effects are contingent on factors like retailer size and type.

Methodology/results: We use restricted-access firm-level microdata from the U.S. Census Bureau's Annual Retail Trade Survey covering approximately 42,500 U.S. retailers and 190,000 firm-year observations from 2006 to 2019. The data uniquely allows us to observe both the physical location of inventory and e-commerce sales shares. Using two-way fixed effects models complemented by robustness analyses, we find that higher SIP is, on average, negatively associated with operating margins. However, this negative relationship is significantly attenuated as ECI increases. Importantly, the moderating effect of ECI is economically large and statistically significant only for large retailers and is strongest for specialty-oriented retailers relative to convenience-oriented ones.

Managerial implications: Our findings show that SIP is not universally beneficial; its profitability implications depend critically on a retailer's channel mix, scale, and type. Accordingly, managers should align inventory placement decisions with their ECI, size, and type. For large retailers with substantial online sales, greater store-based inventory can enhance operating margins, whereas smaller or convenience-oriented retailers may be better served by more centralized inventory holdings.

Keyword: Retail, Store inventory, E-commerce

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1. Introduction

As global e-commerce sales continue to grow and consumer expectations regarding product availability, immediacy, and convenience rise, retailers must manage increasingly complex and costly distribution operations (Caro et al., 2020; McKinley et al., 2025). As a result, the quest to balance efficiency and responsiveness becomes even more critical for retailers' competitiveness and survival. A central element of this balancing act relates to retailers' store inventory positioning (SIP), the share of total inventory held in stores rather than in upstream facilities (e.g., Govindarajan et al., 2021). Inventory positioning involves a fundamental tradeoff between cost efficiency and responsiveness, with upstream inventory enabling risk pooling, and store inventory supporting local availability and fast fulfillment. Understanding how SIP influences retailer performance in a multi-channel setting is critical, especially considering that, while over 96% of the top 500 retail chains have an online presence, brick-and-mortar stores still account for almost 80% of total retail sales in the U.S. (Digital Commerce 360, 2023; LS Retail, 2025).

While prior studies have added to our understanding of how inventory metrics like productivity, leanness, and responsiveness affect firm performance (e.g., Alan et al., 2014; Shockley & Turner, 2015; Chuang et al., 2019), the effects of how the spatial distribution of inventory across retailers' stores and warehouses impacts their performance in a multi-channel context remain largely unexplored. One exception is the work of Qin et al. (2022), who analyze inventory centralization using data from a single e-commerce retailer, but do not consider store inventory or how inventory positioning interacts with retailers' e-commerce intensity (ECI), the share of e-commerce-enabled sales. Seeing that a significant proportion of retailers operate in both physical and online channels (Ratchford et al., 2022), a broader understanding of how SIP and ECI jointly shape retailer profitability is crucial for both theory and practice.

This study explores the relationship between SIP and retailers' operating margins and, importantly, investigates how the nature of this relationship varies with a retailer's ECI. E-commerce has fundamentally transformed the purpose and economics of inventory. Store inventories often double as local e-commerce fulfillment hubs (Gallino & Moreno, 2014) to help meet consumers' higher

expectations with respect to product availability or delivery options and accommodate their changing shopping habits (Pozzi, 2013; Nguyen et al., 2019). These developments imply that the performance effects of SIP are likely to be contingent on retailers' ECI as a primary contextual factor that shapes both the cost and revenue implications of SIP: Higher ECI increases the value of store inventory for fulfillment speed, while also raising the operating costs associated with using stores as decentralized e-commerce fulfillment centers.

Yet, these effects are likely not uniform across retailers. We submit that the effectiveness of SIP and its interaction with ECI depend on retailer size as a key structural firm characteristic. Firm size is a crucial determinant of how effectively and efficiently store inventory can be used in the fulfillment of e-commerce sales: Larger retailers tend to have greater geographic market coverage and greater store density, which enables more efficient ship-from-store operations (Holmes, 2011; Snoeck et al., 2023). Hence, the effect of SIP and its interaction with ECI are expected to vary with retailer size. Taken together, these considerations motivate the following research questions:

(1) How does retailer store inventory positioning (SIP) relate to retailer operating margins?

(2) How does the SIP-operating margins relationship vary as a function of the retailer's e-commerce intensity (ECI)?

(3) How do the effects of SIP and its interaction with ECI on operating margin vary by retailer size?

In addition, we conduct an exploratory analysis examining whether the performance implications of SIP under varying degrees of ECI differ across convenience-oriented and specialty-oriented retailers.

We use restricted-access U.S. Census microdata collected via the Annual Retail Trade Survey (ARTS) to empirically address the research questions outlined above. The ARTS database compiles information on various data items, including some financial and accounting metrics. Two key advantages of this dataset are that it provides both in-store and e-commerce sales information and separately reports store and upstream warehouse inventory data. In addition, the data collection covers a large number of both public and private firms across all retail sectors. This granularity and coverage allow us to observe ECI and SIP directly and to conduct large-scale analyses of their effects on retailer performance. With the

caveat that gaining access to U.S. Census microdata involves a time-consuming application and vetting process and comes with strict physical entry, usage, and disclosure limitations, such data has the potential to spawn future operations management research (Nagaraj et al., 2025), especially since, to the best of our knowledge, its use in prior operations and supply chain management research has been limited.

Our research findings support the central insight that SIP does not have uniformly positive or negative performance implications. Rather, its effect on performance is shaped by retailers' ECI and scale, and it also varies by retailer type (convenience-oriented vs. specialty-oriented). Theoretically, our study extends retail operations and supply chain management research by conceptualizing inventory positioning as a channel-dependent driver of retailer performance. We complement prior retail and inventory literature by showing how changes in the channel mix (brick-and-mortar – e-commerce) alter the strategic role of inventory. Moreover, by examining variations across retailer firm size and type, we document that SIP is not universally effective and depends on key retailer characteristics. Practically, our findings inform retailers' efforts to enhance profitability by aligning inventory strategies with channel mix, size, and type. Empirically, we contribute by leveraging restricted-access U.S. Census microdata that jointly report e-commerce sales and the physical location of inventory for both public and private retailers. This granularity allows us to observe both SIP and ECI directly across the full size distribution of U.S. retailers, extending prior work that has largely relied on publicly listed firms or single-company datasets.

2. Background

2.1. Store inventory positioning and retailer performance

Foundational operations management research demonstrates that inventory positioning is a central determinant of performance in distribution systems. Classic risk pooling theory submits that centralizing inventory upstream reduces inventory costs by smoothing demand variability across locations, whereas decentralizing inventory holdings eliminates these pooling benefits and raises holding and shortage costs (Eppen, 1979). These effects operate primarily on the cost side of retailer performance. Subsequent research generalizes this insight, showing that the cost advantages of centralized inventory persist under various conditions marked by different demand distributions, nonlinear cost structures, and alternative

network configurations (Chen & Lin, 1989; Shen et al., 2003; Yang et al., 2021). Collectively, this body of work establishes that positioning more inventory in stores increases exposure to demand variability and raises inventory-related costs. Upstream centralization, in contrast, enhances risk pooling benefits but may reduce responsiveness, thereby limiting the revenue-side benefits associated with greater product availability and faster fulfillment. This foundational tradeoff provides the theoretical baseline for assessing how SIP shapes retailer performance.

Some research highlights conditions under which decentralized positioning of inventory (e.g., in stores) becomes advantageous. In particular, recent analytical studies show that allocating more inventory to stores improves local availability and can reduce transportation or delivery costs, especially in an omnichannel context where local fulfillment shortens last-mile distances and lead times (Saha & Bhattacharya, 2021; Bayram & Cesaret, 2021; Goedhart et al., 2023; Jiu, 2022). Using store inventory to serve both walk-in and online customers increases the value of store inventory but may also amplify inventory imbalances and compound stockout risks (Acimovic & Graves, 2017; Govindarajan et al., 2021). Likewise, research suggests that customers' valuation of greater responsiveness and faster fulfillment can outweigh the loss of risk pooling benefits associated with decentralization, making downstream inventory holding strategically appealing under certain demand and cost conditions (Yao et al., 2009; Qin et al., 2022). Reinforcing this fundamental tradeoff, network design work concludes that decentralized (store-heavy) inventory improves service and revenue opportunities, whereas centralized (warehouse-heavy) inventory provides stronger protection against demand uncertainty and lower operational costs (Hassanzadeh & Martínez-de-Albéniz, 2026; Lin et al., 2022; Kolyaei et al., 2025).

Taken together, these studies suggest that SIP is a central performance lever, with store-level and upstream inventory each offering distinct cost-side and revenue-side benefits and disadvantages. Conditions that favor greater SIP include environments where responsiveness is important, and stores are used to fulfill online customer demand. These conditions become particularly salient as retailers expand their e-commerce operations, providing strong motivation for examining how ECI moderates the SIP-performance relationship.

2.2. The moderating role of e-commerce intensity on the SIP-performance link

As retailers' e-commerce operations expand, the demands placed on their distribution networks change in ways that shift the optimal inventory positioning (Hübner et al., 2015; Fisher & Raman, 2022), suggesting that the SIP-performance relationship varies with a retailer's ECI. A central effect of greater ECI is that the marginal value of store inventory increases. Store inventory can serve multiple demand streams—walk-in, buy-online-pick-up-in-store, and ship-from-store, for example. This added functionality increases the revenue potential of downstream inventory and strengthens the value of SIP when responsiveness is important (Saha & Bhattacharya, 2021; Bayram & Cesaret, 2021; Goedhart et al., 2023; Jiu, 2022). ECI also heightens customers' sensitivity to availability and fulfillment speed (Nguyen et al., 2019), further amplifying the revenue-side benefits of positioning inventory closer to customers. Empirical evidence shows that shorter fulfillment times and higher logistics performance significantly increase purchase intentions and sales (Deshpande & Pendem, 2023; Baldauf et al., 2024). These findings suggest that the benefits of SIP become stronger as e-commerce demand grows.

At the same time, greater ECI magnifies the operational challenges associated with decentralized inventory. When store inventory serves both in-store and online demand, inventory imbalances and stockout risks become more likely, especially in settings where stores are used for e-commerce order fulfillment (Acimovic & Graves, 2017). Store-based fulfillment also raises picking, packing, and coordination requirements, imposing additional operational burdens compared with upstream fulfillment options (Govindarajan et al., 2021; Goedhart et al., 2023). Thus, even as the revenue benefits of SIP increase with ECI, the operational costs of decentralized inventory also rise.

ECI shifts the core tradeoff considerations between centralized and decentralized inventory. Prior studies incorporating e-commerce fulfillment stress that distribution models with greater SIP gain value as delivery speed becomes more important, while also making retailers more vulnerable to demand variability and store-level handling constraints as e-commerce order volumes grow (Hassanzadeh & Martínez-de-Albéniz, 2026; Lin et al., 2022; Kolyaei et al., 2025). Centralized inventory (i.e., lower SIP), by contrast, retains risk-pooling advantages but does not offer the responsiveness that e-commerce

customers increasingly expect. Accordingly, as ECI rises, the performance consequences of placing inventory predominantly upstream (low SIP) or downstream (high SIP) become more pronounced, increasing the sensitivity of retailer performance to SIP.

Whether ECI strengthens or weakens the SIP–performance relationship depends on whether revenue gains scale more steeply than operational and coordination costs—a question that existing theory does not resolve unambiguously and that we examine empirically.

2.3. Variations in the SIP x ECI interaction by retailer size

We expect that the moderating effect of ECI on the SIP-performance relationship will be stronger for large retailers than for small ones. Large retailers typically operate store networks that are denser and cover larger geographic areas (e.g., Ellickson et al., 2013), thus presenting greater revenue-generating potential for store-based e-commerce order fulfillment. As ECI increases, a retailer with many stores can route most online orders to nearby locations. This creates substantial last-mile cost savings, spreads national e-commerce demand across many inventory points, and delivers consistently faster fulfillment—all advantages that tend to grow disproportionately with the retailer's physical footprint (Govindarajan et al., 2021; Jiu, 2022; Goedhart et al., 2023; Lin et al., 2022). Small retailers with smaller operational networks, on the other hand, have a limited ability to leverage store inventory to generate additional revenue from e-commerce sales.

Moreover, the substantial costs of integrated warehouse management systems, real-time allocation engines, employee training for ship-from-store, and picking/packing infrastructure needed to efficiently and effectively utilize store inventory for both walk-in and e-commerce customer demand confer relative cost advantages on large retailers (Hübner et al., 2015; Fisher & Raman, 2022). Large retailers can amortize these investments over greater volumes and online order quantities. Smaller chains, by contrast, often remain reliant on manual or semi-automated processes (Kembro & Norrman, 2022), which limits both the feasibility and the profitability of greater SIP in a high-ECI context.

Finally, empirical evidence shows that consumers, conditioned by national retailers to expect fast fulfillment, respond more negatively to the typically slower delivery times of smaller retailers (Deshpande

& Pendem, 2023). Hence, high SIP translates into a greater demand uplift for larger retailers than for small ones when ECI is elevated. This suggests that the marginal effect of SIP on performance at high levels of ECI should be significantly greater for large retailers than for small ones.

3. Data and variable measurement

3.1. Data source

This study relies on restricted-use microdata[1] from the U.S. Census Bureau's Annual Retail Trade Survey (ARTS), accessed through the Federal Statistical Research Data Center (FSRDC) program. The ARTS, which transitioned to the Annual Integrated Economic Survey (AIES) in 2024, is an annual survey that compiles detailed financial and operational information from a stratified random sample of public and private retail firms. Critically for our purposes, the ARTS reports (i) total merchandise inventories as well as the dollar value of inventories physically located in retail stores and (ii) total firm sales as well as e-commerce sales.[2] These data enable us to construct measures of SIP and ECI, respectively. Whereas publicly available ARTS data are aggregated at the NAICS-year level, we gained access to firm-year level microdata.

Access to firm-level micro records is highly restricted to researchers whose projects are approved following an extensive application and vetting process by the Census Bureau. Approved researchers must work exclusively within secure FSRDCs that prevent the removal of data from the facility. All output undergoes disclosure avoidance review before release,[3] and researchers are legally prohibited from publishing any statistic that could identify individual establishments or firms. These limitations and safeguards explain why, to the best of our knowledge, no previous published firm-level empirical work has used the SIP or ECI measures available in the ARTS microdata (Nagaraj et al., 2025).

[1] https://www.census.gov/topics/research/guidance/restricted-use-microdata.html.
[2] According to the U.S. Census Bureau, "E-commerce sales are sales of goods and services where the buyer places an order, or the price and terms of the sale are negotiated, over an Internet, mobile device (M-Commerce), extranet, Electronic Data Interchange (EDI) network, electronic mail, or other comparable online system. Payment may or may not be made online." (https://www.census.gov/programs-surveys/arts/about/faq.html).
[3] https://www2.census.gov/adrm/FSRDC/Resources/FSRDC-Disclosure-Avoidance-Methods-Handbook.pdf.

The unique combination of granular inventory-location data and e-commerce sales data, together with the legal and procedural barriers to access, makes the ARTS microdata exceptionally well-suited and effectively irreplaceable for the large-scale testing of how SIP interacts with ECI to affect retailer performance. All analyses reported in this study were conducted in compliance with FSRDC protocols. Descriptive statistics and regression results have been cleared for public release by the Census Bureau's Disclosure Review Board. We combine ARTS microdata with information from the Business Register and the Business Expenses Survey to construct our retailer-year-level dataset.

3.2. Variables

The dependent variable in our research is a retailer's operating margin (*OpMgn*). *OpMgn* is calculated as the ratio of operating income (sales minus COGS and operating expenses) to total revenue and captures firms' operational effectiveness (e.g., Leung et al., 2020; Aral et al., 2023).

The key independent and moderating variables are *SIP* and *ECI*, respectively. *SIP* is measured as the percentage of a retailer's total inventory that is located in stores. We are unaware of any prior research that has deployed such a measure, most likely due to a lack of publicly available data. We do, however, acknowledge potential measurement noise arising from reporting error or heterogeneity in how retailers classify inventory across facilities. To the extent that such random noise is present, it would attenuate estimated coefficients toward zero, biasing against finding statistically significant effects. Hence, we argue that our *SIP* measure will generate conservative estimates. The *ECI* measure is similarly computed as the share of total sales that is e-commerce enabled (Harsha et al., 2019; Milligan et al, 2025).

The control variables include firm size (*FSize*) and firm growth (*FGrowth*). The former is operationalized as the natural logarithm of annual payroll expenses, while the latter is calculated as the year-over-year change in annual payroll (e.g., Guo & Wallskog, 2025). Our baseline two-way fixed

effects model also includes firm and time fixed effects to control for otherwise unobserved time-invariant, firm-specific effects and firm-invariant temporal effects, respectively.[4]

3.3. Sample and descriptive statistics

The estimation sample includes a total of about 190,000 complete firm-year observations pertaining to 42,500 unique retailers across the 2006-2019 time period.[5] A total of 57 five-digit NAICS retail industries are represented in the data.

Table 1 reports descriptive statistics and pairwise correlations for the variables included in the analysis. The average operating margin is 3.5%, with substantial dispersion across firm-year observations. *FSize* and *FGrowth* exhibit considerable variation, consistent with the heterogeneous nature of the sample. *ECI* is fairly low, on average (3.32%), but exhibits significant variation across firms and over time. The average *SIP* of 84.20% suggests that the majority of retailers' inventories are held in stores, on average, but with significant variation (31.32%). Pairwise correlations are generally modest in magnitude, suggesting that multicollinearity is unlikely to be a concern.

Table 1 Descriptive statistics and pairwise correlations (N = 190,000)

Variable	Mean	SD	1	2	3	4
1 OpMgn	0.035	0.168				
2 FSize	6.430	2.257	0.056			
			(0.000)			
3 FGrowth	0.092	0.495	0.001	-0.004		
			(0.712)	(0.112)		
4 ECI	0.033	0.127	0.029	0.076	0.007	
			(0.000)	(0.000)	(0.001)	
5 SIP	0.842	0.313	-0.054	-0.127	-0.002	-0.301
			(0.000)	(0.000)	(0.282)	(0.000)

Notes: P-values of correlations are reported in parentheses. *OpMgn* – operating income over total sales, *FSize* – natural logarithm of annual payroll; *FGrowth* – percent change in annual payroll; *ECI* – percentage of total sales generated via online channels; *SIP* – share of total inventory held in stores. *OpMgn* and *FGrowth* are winsorized at the 1st and 99th percentiles. Source: FSRDC Project Number 1871. Disclosure clearance number CBDRB-FY26-P1871-R12847; CBDRB-FY26-P1871-R12808.

[4] The ARTS collects limited financial information beyond the items listed here, thereby restricting our ability to control for additional firm characteristics. The deployment of firm fixed effects largely overcomes this limitation.
[5] Sample sizes and firm counts are reported as rounded numbers due to disclosure restrictions imposed by the U.S. Census Bureau.

4. Analysis and results

4.1. Two-way fixed effects regression analysis

To examine the relationship between SIP, ECI, and operating margin, we estimate two-way fixed effects regression models, as reported in Table 2. This approach exploits within-firm variation over time while controlling for unobserved, time-invariant retailer characteristics—such as business models, supply-chain design, and managerial practices—that may jointly influence inventory placement, distribution strategy, and profitability. In addition, year fixed effects account for common temporal shocks, including macroeconomic conditions, changes in consumer shopping behavior, and industry-wide trends affecting both online and store-based operations. By focusing on deviations from each firm's own historical baseline and netting out aggregate time effects, the two-way fixed effects specification mitigates concerns related to omitted variable bias and spurious correlation that could arise from persistent cross-sectional differences or common shocks. Compared to random effects models, which require the strong assumption that unobserved firm characteristics are uncorrelated with SIP and ECI, the fixed effects approach provides a more conservative test of the relationships of interest. While this specification does not eliminate all endogeneity concerns, it substantially reduces bias stemming from unobserved heterogeneity and strengthens causal interpretation by relying on within-firm changes in SIP and ECI over time.

Column (1) of Table 2 reports the baseline two-way fixed effects results. Firm size is positively associated with operating margin, reflecting the scale advantages in retail operations. Firm growth, on the other hand, exhibits a negative relationship with operating margins, highlighting the short-term adjustment costs associated with growth. Turning to the primary independent variables, SIP is negatively associated with operating margin. Retailers that hold a larger proportion of inventory in stores experience lower operating performance, on average. This finding is in line with the argument that greater decentralization of inventory holdings leads to the loss of risk pooling benefits and increases in logistics costs. ECI is also negatively related to operating margin. Together, these results suggest that both inventory positioning and e-commerce engagement are systematically related to retailer profitability after accounting for size-related firm characteristics as well as unobserved firm heterogeneity and time effects.

Importantly, the negative main effects of SIP and ECI should not be interpreted as evidence that store-based inventory or e-commerce engagement are intrinsically value-destroying. Rather, these within-firm estimates are consistent with substantial short- to medium-run adjustment costs associated with decentralizing inventory and scaling e-commerce operations.

Table 2 Two-way fixed effects regression results (N = 190,000)

Variables	**(1)**	**(2)**
FSize	0.004***	0.004***
	(0.001)	(0.001)
FGrowth	-0.003***	-0.003***
	(0.001)	(0.001)
SIP	-0.006*	-0.008**
	(0.003)	(0.003)
ECI	-0.034**	-0.052***
	(0.011)	(0.014)
SIP × ECI		0.034*
		(0.014)
Constant	0.014†	0.016*
	(0.008)	(0.008)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R^2	0.606	0.606
F-statistic	9.073***	8.337***

Notes: DV = OpMgn; robust standard errors are reported in parentheses. Source: FSRDC Project Number 1871. Disclosure clearance number CBDRB-FY26-P1871-R12847; CBDRB-FY26-P1871-R12808. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Column (2) extends the baseline specification by introducing the interaction between SIP and ECI. The associated coefficient estimate is positive and statistically significant, indicating that the negative association between SIP and operating margin is attenuated for firms that generate a relatively greater share of e-commerce sales. At the same time, the main effects of SIP and ECI remain negative. This pattern suggests that while higher SIP and ECI are each associated with lower operating margins, their joint presence partially offsets these effects. Overall, the results are consistent with the view that ECI moderates the profitability implications of SIP, highlighting the importance of aligning inventory placement with distribution strategy.

4.2. Differences by retailer firm size

As discussed in Section 2.3, we investigate whether the moderating effect of ECI on the relationship between SIP and operating margin varies systematically with retailer firm size. Specifically, we expect that the SIP × ECI interaction will be stronger for larger retailers than for smaller ones. To examine this possibility, we conduct a subsample analysis in which we divide retailers into quartiles based on the number of employees as a proxy for retailers' network size and re-estimate our two-way fixed effects model separately for each group. We use employee counts rather than payroll expenses because the number of employees more closely reflects the scale of a retailer's operational footprint, which is key when assessing the ability to leverage store inventory for e-commerce fulfillment. The quartile regression results are reported in Table 3 below.

Table 3 Two-way fixed effects regression results by quartiles of retailer firm size

Variables	**1st quartile**	**2nd quartile**	**3rd quartile**	**4th quartile**
FSize	0.001	0.004	0.013**	0.000
	(0.002)	(0.004)	(0.004)	(0.003)
FGrowth	−0.004**	−0.003	0.000	−0.002
	(0.001)	(0.002)	(0.002)	(0.002)
SIP	0.002	0.002	−0.006	−0.024***
	(0.008)	(0.006)	(0.005)	(0.006)
ECI	−0.015	−0.064†	−0.049†	−0.080**
	(0.032)	(0.036)	(0.027)	(0.024)
SIP × ECI	0.017	0.023	0.028	0.064**
	(0.035)	(0.028)	(0.028)	(0.024)
Constant	0.015	0.015	−0.044	0.062*
	(0.012)	(0.025)	(0.029)	(0.024)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
No. of observations	46,000	45,000	45,500	47,000
No. of firms	12,500	12,000	11,500	9,000
F-statistic	2.074†	1.196	3.109**	4.664***
R^2	0.669	0.638	0.592	0.549

Notes: DV = OpMgn; robust standard errors are reported in parentheses. Source: FSRDC Project Number 1871. Disclosure clearance number CBDRB-FY26-P1871-R12847; CBDRB-FY26-P1871-R12808. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The results reveal a clear pattern across firm size quartiles. For the smallest retailers (Quartiles 1 and 2), the interaction between SIP and ECI is positive but statistically insignificant, indicating that higher ECI does not materially alter the performance implications of SIP for these firms. For mid-sized retailers (Quartile 3), the interaction is larger in magnitude but still statistically insignificant. In contrast, for the largest retailers (Quartile 4), the SIP × ECI interaction is positive and statistically significant. This pattern suggests that the performance benefits of aligning store inventory with ECI are concentrated among retailers with the scale and store density needed to effectively leverage stores for the fulfillment of e-commerce demand. In other words, e-commerce enhances the value of store inventory primarily for large retailers, while smaller retailers are less able to convert higher ECI into improved margins through SIP. These results indicate that the profitability consequences of SIP under varying degrees of ECI are fundamentally shaped by retailers' ability to leverage their operational scale, rather than by ECI alone.

4.3. Supplemental analysis: Convenience-oriented versus specialty-oriented retailers

Retailers differ not only in size but also in terms of the primary consumer needs they serve (e.g., Bucklin, 1963; Kotler & Keller, 2016). Convenience-oriented retailers (e.g., grocery and convenience stores), for example, predominantly sell frequently purchased, lower-priced items that consumers buy with minimal search effort and low involvement (Thirumalai & Sinha, 2005). Specialty-oriented retailers (e.g., new car dealers and furniture stores), on the other hand, focus on branded or unique items for which consumers exhibit strong preferences and are willing to exert considerable effort to acquire (Thirumalai & Sinha, 2005). These retailer types serve as proxies for systematic differences in value, demand variability, and consumers' sensitivity to fulfillment speed.

We expect that the positive moderating effect of ECI on the SIP–performance relationship will be stronger for specialty-oriented retailers as compared to convenience-oriented retailers. When ECI is high, the revenue premium from faster fulfillment enabled by greater SIP is substantially larger for specialty goods because customers exhibit high service-time elasticity of demand (Qin et al., 2022; Deshpande & Pendem, 2023). High SIP enables specialty-oriented retailers to convert this elasticity into higher sales

and margins by offering same-day or next-day delivery and in-store pickup options that e-commerce customers value. Conversely, the customers of convenience-oriented retailers are typically less sensitive to delivery times because their purchases are habitual and replenishment-driven (Goedhart et al., 2023; Hübner et al., 2015). In such instances, SIP yields limited benefits while amplifying operational burdens such as picking complexity, shrinkage, and stockout spillovers across channels that come with greater ECI (Acimovic & Graves, 2017; Govindarajan et al., 2021). Consequently, the net benefit of high SIP under rising ECI should be greater for specialty-oriented retailers than for convenience-oriented retailers.

We use 2017 industry-level inventory turnover data as a proxy to classify retail industries as convenience-oriented and specialty-oriented, respectively. Inventory turnover varies systematically across types of retail operations, with convenience-oriented retailers typically exhibiting high inventory turnover, and retailers focused on highly differentiated or slower-moving specialty goods having lower turnover ratios (e.g., Gaur et al., 2005). The following retail segments are included in the convenience-oriented retailer category: Supermarkets (NAICS 44511), Convenience Stores (NAICS 44512), Beer, Wine, and Liquor Stores (NAICS 44531), and Pharmacies and Drug Stores (NAICS 44611)—all with inventory turns of nine or greater in the 2017 reference year. In turn, New Car Dealers (NAICS 44111), Furniture Stores (NAICS 44211), Optical Goods Stores (NAICS 44613), Jewelry Stores (NAICS 44831), and Musical Instrument Stores (NAICS 45114) all turned their inventory fewer than three times in 2017 and are included in the specialty-oriented retailer group. While our classification operates at the retail industry level, it reflects the dominant consumer needs retailers serve and captures persistent differences in retail operating environments that shape the economics of inventory positioning.

Table 4 reports the estimation results for the convenience-oriented and specialty-oriented retailer groups. In line with the expectations outlined above, the results are substantially stronger for the latter than the former. For specialty-oriented retailers, higher ECI significantly increases the marginal value of SIP, consistent with the idea that responsiveness-driven revenue gains from store-based fulfillment are particularly salient in settings characterized by lower purchase frequency and greater customer sensitivity to availability and fulfillment speed. By contrast, we find no comparable evidence for convenience-

oriented retailers, suggesting that the cost-side burdens of decentralized inventory—such as inventory imbalances and store-level handling requirements—are less likely to be offset by incremental revenue gains in high-turnover, price-competitive retail formats. Taken together, these findings indicate that the performance implications of SIP under rising ECI are highly contingent on retailers' underlying consumer demand characteristics.

Table 4 Two-way fixed effects regression results for convenience-oriented retailers versus specialty-oriented retailers

	Convenience-oriented retailers (N=25,000)		**Specialty-oriented retailers (N=35,500)**	
Variables	**(1)**	**(2)**	**(3)**	**(4)**
FSize	0.003	0.003	0.004	0.004
	(0.003)	(0.003)	(0.003)	(0.003)
FGrowth	-0.004**	-0.004**	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)
SIP	0.006	0.007	-0.016†	-0.019*
	(0.010)	(0.010)	(0.009)	(0.009)
ECI	0.0257	0.064	-0.025	-0.159*
	(0.044)	(0.078)	(0.029)	(0.063)
SIP × ECI		-0.050		0.141*
		(0.070)		(0.063)
Constant	0.001	-0.001	0.019	0.021
	(0.019)	(0.019)	(0.024)	(0.024)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
R^2	0.640	0.640	0.538	0.538*
F	1.930	1.614	1.563	2.228

Notes: DV = OpMgn; robust standard errors are reported in parentheses. Source: FSRDC Project Number 1871. Disclosure clearance number CBDRB-FY26-P1871-R12847; CBDRB-FY26-P1871-R12808. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

4.4. Robustness check: Control function analysis

Although the two-way fixed effects specification mitigates many sources of bias by absorbing time-invariant firm heterogeneity and common shocks, endogeneity concerns may persist if unobserved, time-varying factors jointly influence the outcome and any of the independent or moderating variables. In

addition, reverse causality represents a plausible concern: firms experiencing changes in operating performance may subsequently adjust their e-commerce engagement. For example, retailers may expand such engagement in response to margin pressure or by scaling it back following weak performance. To address concerns regarding the potential endogeneity of ECI, in particular, we complement the two-way fixed effects analyses with a control function approach. This framework is particularly suitable in our setting because the potentially endogenous variable, ECI, enters the model both directly and through an interaction with SIP (Villena & Dhanorkar, 2020). The control function approach consists of a first stage where we regress ECI on all exogenous variables and instruments that are both relevant and plausibly exogenous sources of variation in retailers' ECI. In the second stage, we then include the resulting residual in the outcome equation, thereby testing whether the aforementioned endogeneity concerns, if they exist, materially affect the estimation results. In this sense, the control function analysis provides a conservative robustness check that complements (rather than replaces) our primary identification based on within-firm variation over time.

We deploy two instruments in the first-stage regression analysis: lagged peer e-commerce adoption (*PeerECAdopt_lagged*) and lagged spending on data and computer services as a share of total operating expenses (*D&CSvcsShare_lagged*). *PeerECAdopt* is measured as the percentage of peer firms in the same five-digit NAICS industry that are engaged in e-commerce, where a firm's e-commerce engagement is coded as 0 or 1 depending on whether or not a firm engages in e-commerce sales, and is intended to capture diffusion and competitive imitation dynamics. Firms operating in retail industries with higher prior e-commerce penetration may face stronger competitive pressure or have greater learning opportunities, increasing their own e-commerce engagement. The lagged share of data and computer services expenses, in turn, is a proxy for prior digital infrastructure and IT service intensity, which should be predictive of a firm's capability to expand e-commerce operations. We acknowledge that neither instrument is beyond critique. Peer adoption could be correlated with common demand or competitive shocks, and digital spending may affect profitability through channels other than ECI. We, therefore, view

these instruments as providing a reasonable (but not definitive) basis for identification, and we interpret the control function results as a robustness assessment rather than as a primary identification strategy.

The sample size decreases substantially in the control function analysis because data and computer services expense information is not available for all firm-years. Specifically, these data are obtained from the Business Expenses Survey collected in Census years (2007, 2012, and 2017) and are not consistently available for all retailers. To construct firm-year measures, we interpolate values between Census years (2007, 2012, 2017) and extrapolate backward to 2006 and forward to 2018 and 2019 using Stata's *ipolate* function.

The control function regression results are shown in Table 5. The first-stage results indicate that both instruments are strongly related to ECI. *PeerECAdopt_lagged* is positively and statistically significantly associated with focal firms' ECI, consistent with diffusion, learning, and competitive imitation mechanisms. The lagged share of expenses related to data and computer services (*D&CSvcsShare_lagged*) also carries a statistically significant, albeit unexpectedly negative coefficient estimate. While digital expenditures may facilitate online sales, they encompass a broad set of IT-related activities, many of which support internal operations, legacy systems, or store-based processes rather than customer-facing e-commerce channels. As a result, higher spending on such services may reflect alternative digital investment paths or substitution away from direct e-commerce engagement. We also observe that SIP is negatively associated with ECI, consistent with firms placing relatively more inventory in stores when they have a lower share of online sales. Overall, the first-stage model is jointly significant.

Several diagnostic statistics provide evidence of the appropriateness of our identification strategy. The Cragg–Donald Wald F statistic (126.5) indicates strong instrument relevance under the assumption of homoskedasticity. The Kleibergen–Paap rk Wald F test is robust to violations of this assumption, and the test statistic of 7.388 provides evidence of adequate instrument relevance. Finally, the Hansen J test does not reject the joint validity of the overidentifying restrictions ($J = 0.434$, $p = 0.5099$), which is consistent

with instrument exogeneity. Taken together, these diagnostics suggest that the instruments are relevant and that the overidentifying restrictions are not violated.

Table 5 Control function regression results (N = 54,500)

	(1) 1st stage (DV: ECI)	(2) 2nd stage (DV: OpMgn)	(3) 2nd stage (DV: OpMgn)
FSize	0.002	0.001	0.002
	(0.002)	(0.004)	(0.004)
FGrowth	-0.006***	0.007†	0.007†
	(0.002)	(0.004)	(0.004)
SIP	-0.023***	-0.026**	-0.035***
	(0.006)	(0.011)	(0.011)
PeerECAdopt_lagged	0.099***		
	(0.025)		
D&CSvcsShare_lagged	-0.039**		
	(0.015)		
Residual (control function)		0.271	0.243
		(0.264)	(0.258)
ECI		-0.318	-0.352
		(0.263)	(0.260)
SIP x ECI			0.122***
			(0.037)
Constant	0.017	0.064*	0.070*
	(0.015)	(0.031)	(0.031)
Firm fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
R^2	0.281	0.001	0.000
F-statistic	8.738***	2.036**	2.347**

Notes: Robust standard errors are reported in parentheses. Source: FSRDC Project Number 1871. Disclosure clearance number CBDRB-FY26-P1871-R12847; CBDRB-FY26-P1871-R12808.
† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The second-stage control-function results are broadly consistent with the two-way fixed effects regression results. In particular, SIP remains negatively associated with operating margin, and the interaction between SIP and ECI is positive and statistically significant, indicating that the negative effect of higher SIP on operating margins is attenuated when a retailer's ECI is greater. The residual (control function) term is not statistically significant in either specification of the second-stage model, suggesting limited evidence that endogeneity in ECI materially biases the estimated relationships of interest. Still, we

acknowledge that the instruments are imperfect and rely on exclusion restrictions that cannot be tested directly. Accordingly, our primary identification continues to stem from within-firm variation over time in SIP and ECI, with the control function results providing complementary support for the robustness of the central inferences.

5. Discussion and conclusion

5.1. Summary of findings

This study examines how SIP relates to retailer profitability and how this relationship depends on retailer ECI, potentially varying with retailer size and type (convenience-oriented vs. specialty-oriented). Using U.S. Census microdata that report both the physical location of inventory and e-commerce sales, we find that, on average, a higher share of inventory held in stores is associated with lower operating margins. This baseline effect is consistent with classical inventory theory emphasizing the loss of risk-pooling benefits and higher operational costs associated with decentralized inventory (Eppen, 1979). We also find that greater ECI is, on average, negatively related to operating margins. These negative average associations likely reflect transitional and coordination costs rather than long-run value destruction. Many retailers may rationally accept short-term margin pressure when expanding e-commerce or repositioning inventory in order to enhance service levels, defend market share, or build omnichannel capabilities that pay off over longer horizons.

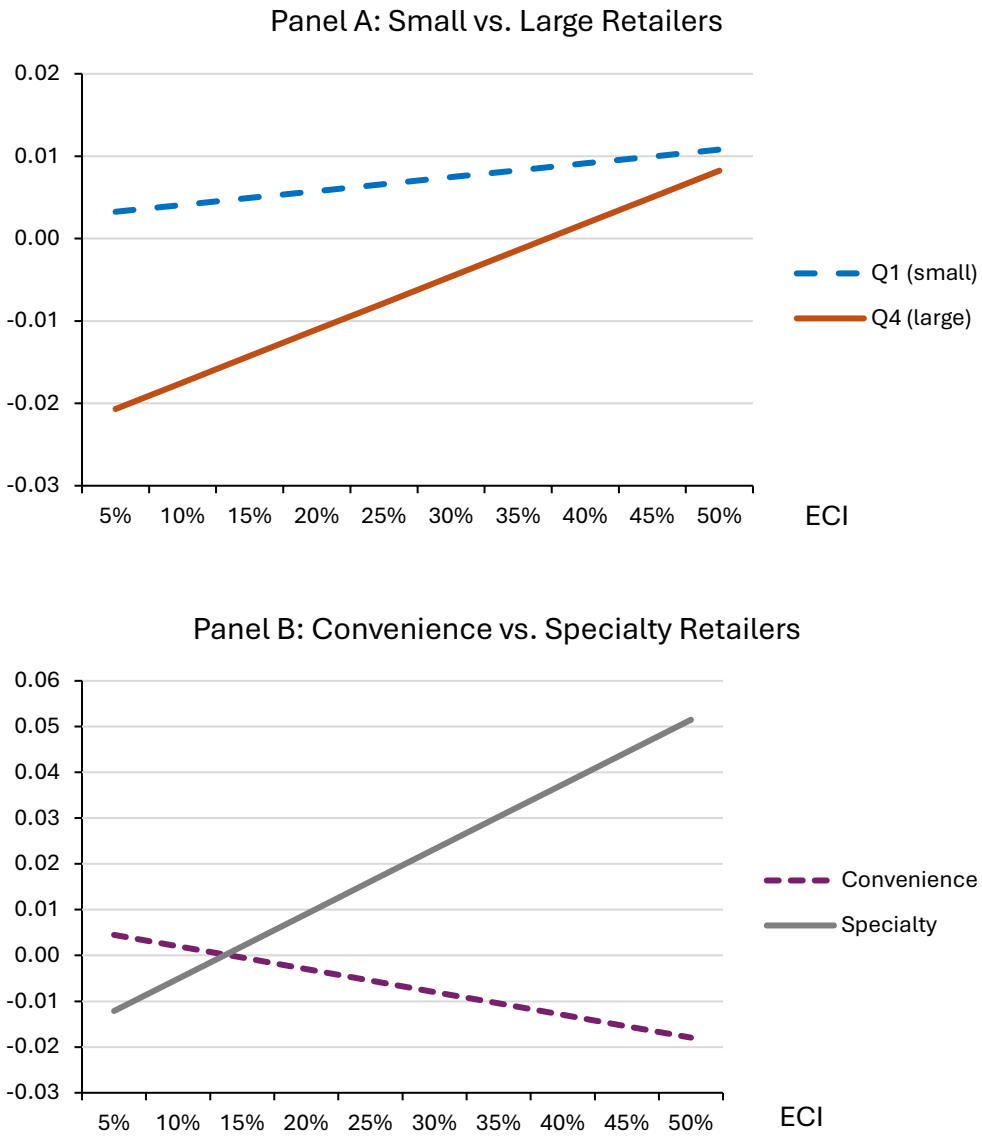
Importantly, our results show that SIP and ECI interact in shaping retailer performance. The negative association between SIP and operating margins is significantly attenuated as ECI increases, indicating that store-based inventory becomes more valuable when it can be leveraged to support faster online fulfillment. This moderating effect, however, is highly heterogeneous. It is statistically significant only for large retailers, consistent with their ability to exploit dense operational networks and their often superior technological and operational infrastructure. To illustrate the economic magnitude of the observed effects, consider a large retailer with a low ECI value of 1%. The marginal effect of a 10% increase in SIP on this retailer's operating margin will be −0.00234 (=0.1*[−0.024+0.064*0.01]), which equates to about a decrease of 6.7% relative to the overall average of retailers' operating margins (3.5%).

Now consider that same retailer at a point in time where its ECI is 20%. In this case, the marginal effect of the same 10% increase in SIP will be -0.0018, equivalent to a 5% decrease relative to the average operating margin. In other words, increasing ECI from 1% to 20% reduces the marginal cost of SIP by more than 25%, illustrating how e-commerce transforms the economic role of store inventory. Supplemental analyses further reveal that this positive interaction is concentrated among specialty-oriented retailers, for whom customers place greater value on availability and fulfillment speed, whereas no comparable benefit emerges for convenience-oriented retailers.

To further illustrate the results, Figure 1 plots the marginal effects of SIP on operating margins across different levels of ECI, separately for small vs. large retailers and for convenience-oriented vs. specialty-oriented retailers. Turning to Panel A first, we observe that the marginal effect of SIP generally increases with ECI, indicating that store-based inventory becomes more valuable as online sales grow. However, the magnitude of these effects differs by retailer size: For large retailers, the marginal effect of SIP is strongly negative at low levels of ECI, reflecting the substantial coordination and handling costs associated with decentralized inventory. As ECI increases, these costs are increasingly offset by the ability to leverage store inventory for e-commerce order fulfillment, resulting in rising and, eventually, positive marginal effects. In contrast, for small retailers, the marginal effect of SIP is positive even at low ECI levels and increases only modestly as ECI rises, suggesting more limited potential for scale-driven benefits. Panel B depicts the contrasting marginal effects for convenience-oriented and specialty-oriented retailers. For the latter, the marginal effect of SIP increases steeply with ECI, becoming strongly positive at moderate-to-high levels of ECI. This pattern suggests that, as e-commerce expands, store inventory becomes an increasingly valuable asset for specialty retailers by enabling fast, localized fulfillment and supporting high-margin, differentiated assortments. In contrast, for convenience retailers, the marginal effect of SIP declines slightly as ECI increases and turns negative at higher levels of online sales. This indicates that, for retailers focused primarily on frequently purchased commodity items, the operational and coordination costs of store-based fulfillment outweigh the responsiveness benefits as e-commerce

grows. Collectively, these findings demonstrate that the performance implications of SIP are contingent on channel mix, scale, and retailer type rather than universal across retail settings.

Figure 1 Marginal effects of SIP conditioned on ECI and varying by Retailer Size and Type



5.2. Contributions to theory and literature

This study makes several contributions to the operations management, retail, and omnichannel logistics literatures. First, we conceptualize SIP as a channel-dependent performance driver rather than a static structural choice. While classical inventory theory emphasizes the cost advantages of upstream inventory centralization through risk pooling (Eppen, 1979; Chen & Lin, 1989; Shen et al., 2003), our results

demonstrate that the profitability implications of SIP depend systematically on retailers' ECI. By empirically documenting that the negative performance effects of decentralized (store-based) inventory are attenuated as ECI increases, we provide large-scale field evidence that the cost–responsiveness tradeoff shifts as the channel mix evolves. This extends analytical and single-firm studies of inventory positioning (Yao et al., 2009; Qin et al., 2022) by showing that the value of downstream inventory is contingent on the extent to which retailers serve demand across physical and online channels.

Second, we contribute to the growing omnichannel operations literature by highlighting the role of inventory location as a key mechanism through which online and offline channels interact. Prior research has primarily focused on fulfillment policies, channel integration, and delivery speed (Gallino & Moreno, 2014; Gao & Su, 2017; Fisher et al., 2019), often abstracting away from where inventory is physically held. Our findings show that e-commerce does not uniformly favor centralized or decentralized inventory structures; instead, it reshapes the marginal value of store inventory in ways that depend on operational scale and consumer demand characteristics (as approximated by retailer type). By documenting a positive SIP x ECI interaction for large and specialty-oriented retailers, we advance a contingency perspective on omnichannel retailing that helps reconcile mixed theoretical predictions in recent analytical and network design work (Bayram & Cesaret, 2021; Saha & Bhattacharya, 2021; Lin et al., 2022; Goedhart et al., 2023).

Third, we contribute to empirical retail operations research by documenting systematic heterogeneity in the SIP–performance relationship across retailer size and retailer types. We show that the benefits of aligning store inventory with e-commerce demand accrue primarily to large retailers with dense networks (Holmes, 2011; Ellickson et al., 2013) and to specialty-oriented retailers serving customers with high service-time sensitivity (Nguyen et al., 2019; Deshpande & Pendem, 2023). This finding helps explain why omnichannel strategies that emphasize store-based fulfillment succeed for some retailers but fail for others, and it challenges implicit assumptions in the literature that omnichannel capabilities scale linearly across firms (Hübner et al., 2015; Fisher & Raman, 2022).

Finally, we make a methodological contribution by leveraging restricted-access U.S. Census microdata that jointly report e-commerce sales and the physical location of inventory for both public and private retailers. This unique data allows us to directly measure SIP and ECI at scale. In doing so, we demonstrate the value of Census microdata for advancing empirical operations management research (Nagaraj et al., 2025) and open new avenues for studying inventory, fulfillment, and channel strategy in multi-channel retail environments.

5.3. Managerial implications

This study offers several actionable insights for retail managers and decision makers. Our findings caution against one-size-fits-all approaches to store-based inventory positioning. On average, holding a larger share of inventory in stores is associated with lower operating margins, underscoring the cost penalties of decentralized inventory holdings. SIP must be carefully evaluated in light of the mix of retailers' sales across store and online channels. Indeed, our results highlight ECI as a critical contextual factor that shapes the value of store inventory. While greater ECI is itself associated with margin pressure, it also increases the marginal value of store inventory by enabling faster fulfillment and greater local availability. Importantly, these benefits do not materialize uniformly: Only larger retailers with sufficiently dense store networks can effectively leverage store inventory to support e-commerce sales. Hence, retailers must make SIP decisions that align with their degree of ECI and are commensurate with their network size. Our findings with respect to variations in the effects of SIP and ECI across convenience-oriented versus specialty-oriented retailers are also insightful. Managers should align inventory positioning and investment decisions not only with firm size and channel mix but also with their customers' needs and expectations. For specialty retailers, investments that enable stores to support online fulfillment may improve performance, whereas convenience-oriented retailers may achieve better outcomes by maintaining more centralized inventory structures. Overall, our results suggest that effective omnichannel inventory strategies require careful alignment between inventory positioning, e-commerce engagement, firm scale, and retailer type. Rather than treating stores as default fulfillment nodes, managers should view store inventory as a strategic asset whose value depends on the retailer's ability to

leverage it to better serve e-commerce customers without incurring disproportionate costs. Put differently, a key diagnostic question for managers is not whether store-based fulfillment is feasible, but whether their store network density and operational systems are sufficient to convert store inventory into faster, more reliable e-commerce fulfillment without incurring excessive handling and coordination costs.

5.4. Limitations and future research opportunities

This study has some limitations that suggest promising directions for future research. First, while our use of two-way fixed effects and control function analyses mitigates key endogeneity concerns, we cannot fully rule out the influence of unobserved, time-varying factors that jointly affect SIP, ECI, and performance. However, the consistency of our findings across specifications, subsamples, and robustness checks provides confidence in the directional nature of the documented relationships. Future research could complement our approach by exploiting quasi-experimental shocks—such as changes in (state-specific) fulfillment regulations, for example—that differentially affect retailers' ability to deploy store-based inventory in support of e-commerce demand. Second, our measures of SIP and ECI, while uniquely direct, are aggregated at the firm-year level. This prevents us from observing how inventory positioning varies across individual stores, regions, or product categories within firms, and how such heterogeneity affects local fulfillment performance. While this aggregation does not bias our estimates of firm-level profitability effects, it limits insight into the micro-level mechanisms through which SIP creates value. To address this shortcoming, future work could integrate Census microdata with establishment-level data or proprietary store-level datasets.

Beyond these limitations, our study opens several broader avenues for future research. Substantively, researchers could extend our framework by examining how SIP and ECI interact with other strategic levers such as pricing, assortment breadth, fulfillment guarantees, and investments in automation, or by exploring dynamic adjustments in inventory positioning as retailers scale their e-commerce operations over time. Methodologically, our work highlights the considerable but underutilized potential of restricted-access U.S. Census microdata for operations management research. Future studies could leverage these data to examine topics such as capacity utilization, labor–inventory tradeoffs,

technology adoption, and supply chain resilience across a wide range of industry settings. More broadly, increased engagement with Census microdata can enable research that is simultaneously large-scale, granular, and representative, thereby complementing analytical, case-based, or empirical research that is limited in scope.

5.5. Conclusion

This study examines how inventory positioning shapes retailer performance in retail settings marked by varying degrees of e-commerce engagement. Using U.S. Census microdata that directly capture inventory location and e-commerce sales, we show that the performance implications of store-based inventory depend on retailers' e-commerce intensity, scale, and type. Overall, our findings position inventory location as a context-dependent strategic lever whose value is shaped by ECI and scale, offering large-scale empirical evidence on how e-commerce demand fundamentally alters classic inventory management tradeoffs.

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